Experiment 6 Date:

Experiment with different layouts and color schemes for an app.

Collect user feedback on aesthetics and usability using GIMP (GNU Image Manipulation Program)

Name: Jhasmitha D **Roll number:** 240701216

Aim:

The aim of this experiment is to trial different app layouts and color schemes for a mobile application prototype using GIMP (GNU Image Manipulation Program) and to evaluate user feedback on aesthetics and usability.

Procedure:

1. Install GIMP

- Downloaded GIMP from the official website: https://www.gimp.org/
- Installed it on the computer and launched the application.

2. Create a New Project

- Opened GIMP and navigated to File > New to create a new project.
- Set canvas dimensions to 1080 x 1920 pixels for a standard mobile portrait screen.

3. Design the Base Layout

- Used the Rectangle Select Tool to define sections: Header, Content Area, and Footer.
- Filled sections with base colors using the Bucket Fill Tool.
- Added text labels, buttons, and input fields using the Text Tool and Shape Tools.
- Organized all UI elements in separate named layers (e.g., Header, Button1, InputField).

4. Experiment with Color Schemes

- Duplicated the base layout three times via right-click > Duplicate Image.
- Applied different color schemes to each duplicate using the Colorize Tool and Bucket Fill Tool.
- Exported each variant via File > Export As: Layout_A.png, Layout_B.png, Layout_C.png.

Layout Variants Created:

- Layout A – Minimal Dark Theme: Dark navy background, red-pink accent, white text.
- Layout B – Blue and White: White background, dark blue primary, mid-blue accent.

- Layout C – Green and Gray: Light gray background, forest green primary, soft blue accent.

5. Collect User Feedback

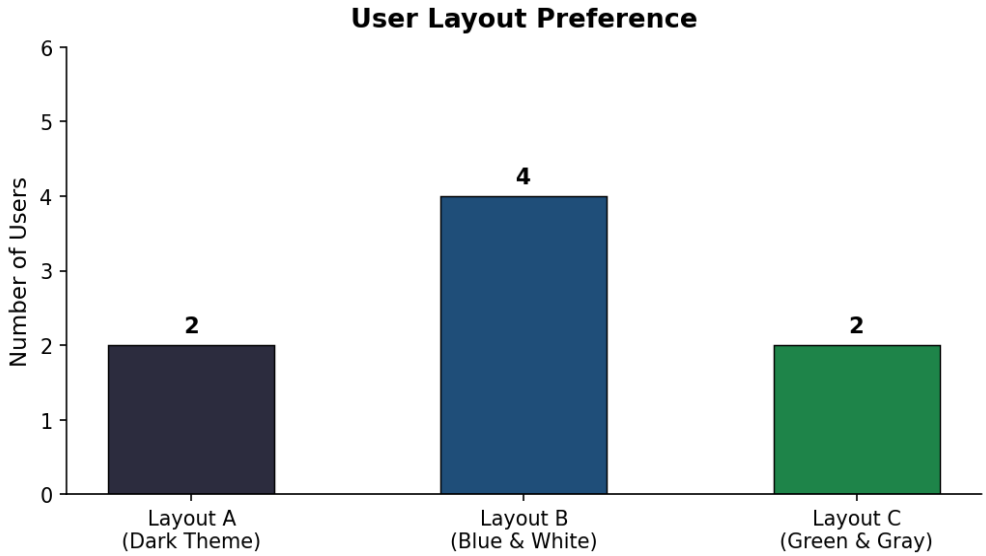
- Created a Google Form with questions on usability and aesthetics rated on a scale of 1 to 5.
- Shared exported PNG files of all three layouts with 8 respondents.
- Collected responses on layout preference, color scheme preference, and overall ratings.

6. Iterate and Refine

- Layout B (Blue and White) was identified as most preferred based on feedback.
- Refinements made: increased button size, improved header contrast, adjusted font weight.
- Conducted a final review to confirm usability and aesthetic improvements.

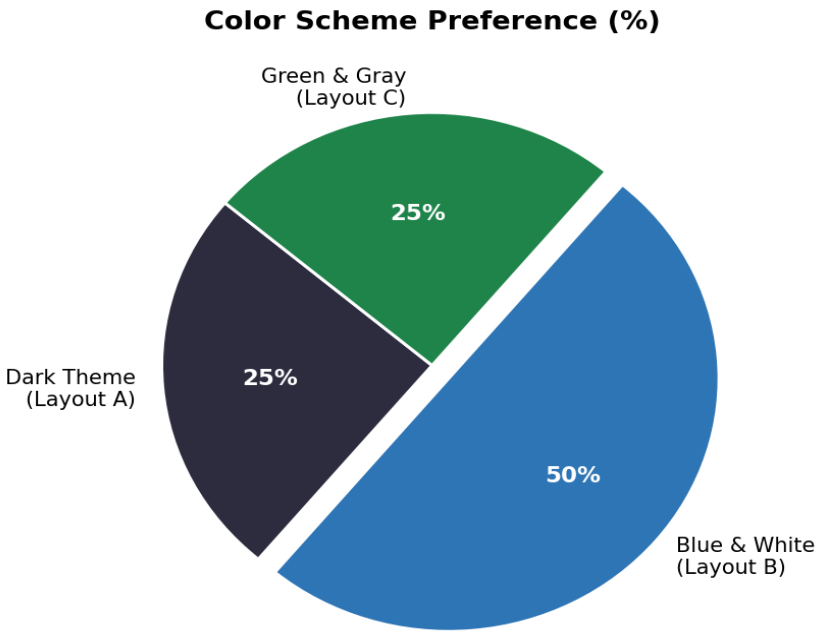
Output:

Graph 1: User Layout Preference



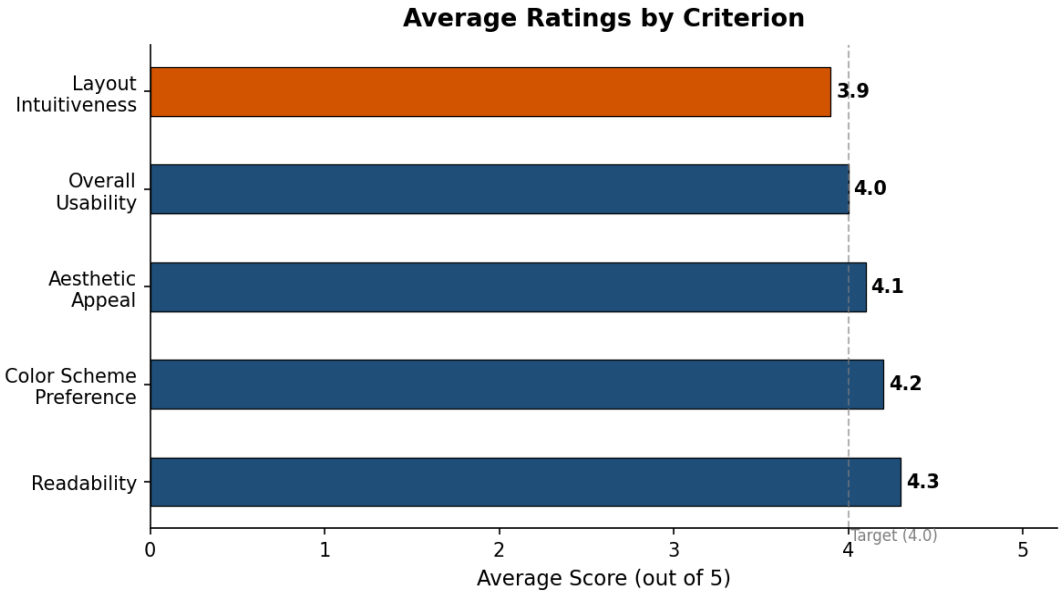
The bar chart shows the number of users who preferred each layout variant. Layout B received the highest votes (4 out of 8 users), followed by Layout A and Layout C with 2 votes each.

Graph 2: Color Scheme Preference



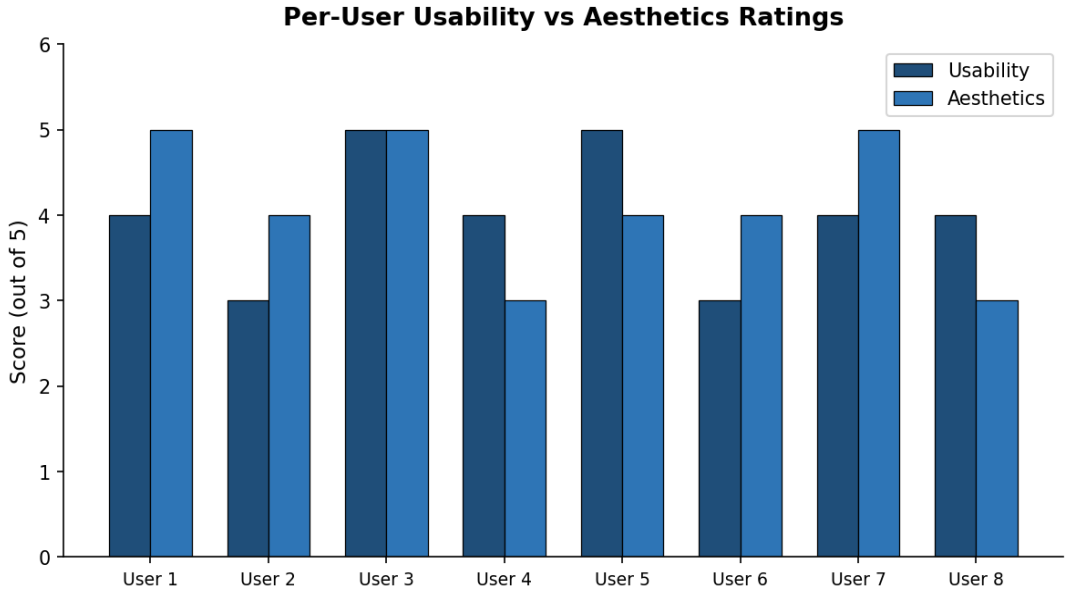
The pie chart shows color scheme preference distribution. Blue and White (Layout B) was preferred by 50% of respondents, while Dark Theme and Green and Gray were each preferred by 25%.

Graph 3: Average Ratings by Criterion



The horizontal bar chart shows average user ratings (out of 5) for five criteria. Readability scored highest at 4.3 and Layout Intuitiveness scored lowest at 3.9, indicating an area for further improvement.

Graph 4: Per-User Usability vs Aesthetics Ratings



The grouped bar chart compares individual user ratings for Usability and Aesthetics. Most users rated Layout B highly on both parameters, with Usability averaging 4.0 and Aesthetics averaging 4.1 overall.

Result:

The experiment was successfully completed. Three app layout variants were designed in GIMP with different color schemes. User feedback was collected from 8 respondents via Google Forms. Layout B (Blue and White) was the most preferred layout with an average usability rating of 4.0 / 5 and an aesthetics rating of 4.1 / 5. The survey graphs confirm that a clean, high-contrast color scheme significantly improves both the visual appeal and usability of an application interface.